

Project Title:**Brute-Force Attack Simulation on SSH and RDP Services Using Hydra and Metasploit in a Dockerized Lab****Summary:**

This project simulates brute-force attacks against both SSH and RDP services within a controlled Docker environment to demonstrate the risks of weak credentials and unpatched systems. It uses Ansible to automate deployment of a Kali Linux attacker, an Ubuntu SSH target, and a Windows RDP target, all networked internally via Docker.

From the Kali container, **nmap** is used to identify open ports (22/3389), **hydra** brute-forces weak credentials (**toor**), and **metasploit** scans the Windows target for the BlueKeep (CVE-2019-0708) vulnerability. A Python script optionally automates these steps. The results highlight how attackers can exploit default or weak passwords and outdated services, reinforcing the importance of system hardening and patch management.

Devices and Technologies:

- **Ansible** – Automates setup and provisioning of all lab containers and required tools.
- **Python** - scripts would be launch from Kali machine to Linux and Windows server containers
- **Docker & Docker Compose** – To isolate and orchestrate attacker and target containers in a repeatable environment.
- **Kali Linux Container** – Used as the attacking machine, equipped with Hydra, Nmap, Metasploit, and supporting tools.
- **Ubuntu-based SSH Target Container** – Hosts an SSH service with a weak password.
- **Windows-based RDP Target Container** – Simulates a Windows environment running Remote Desktop Protocol (RDP) with weak credentials and a vulnerable configuration.
- **Hydra** – Used to brute-force both SSH and RDP login credentials.
- **Metasploit** – Used to scan the Windows container for the BlueKeep RDP vulnerability.
- **Colima (on macOS)** – Provides Docker backend support on macOS.

Key Tools: Docker, **Ansible**, Kali Linux, Linux/Windows server container, Hydra, Metasploit, Nmap, **Python**, Colima/Docker (macOS)